QP Code: 25901855

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

SECOND SEMESTER

Skill Development Components (SDC) – MG2SDCFTA100 – FOOD PRESERVATION

(2025 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I), and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or Two Sentences

Answer any **10** questions

Each question carries **2** Marks

1. **List any two purposes of blanching.** **[U] / [CO1]**
   Enzyme inactivation, cleaning, microbial reduction.

2. **Lits the extrinsic factors that affects the growth of microbes in foods.** **[K] / [CO1]**
   Temperature, Oxygen availability, Relative humidity.

3. **Define semi-perishable foods with example.** **[K] / [CO1]**
   Semi-perishable foods are foods that have a moderate shelf life—they do not spoil as quickly as perishable foods, a few weeks to a month. E.g. Potato, Onion, apple etc.

4. **Recognize how foods get freezed during Immersion freezing.** **[K] / [CO2]**
   Immersion freezing is a rapid freezing method in which food is directly submerged in a very cold liquid refrigerant.

5. **Define dehydration process in foods. Mention the moisture percentage of dehydrated foods.** **[K] / [CO2]**
   Removal of moisture from food using controlled, artificial methods. 10-15%.

6. **Define the principle of drying foods in a Freeze Dryer.** **[U] / [CO2]**
   Sublimation.

7. **Differentiate between contact type and non-contact type heat exchangers.**
[U] / [CO2]

In contact type, the food and the heating medium will be in direct contact. But in non-contact type, their will be a barrier between heating medium and food.

8. **State the principle of low temperature preservation of foods. [U] / [CO2]**

For the growth and multiplication of spoilage microorganisms, it requires an optimum temperature. In refrigeration and freezing since the temperature is too low, the microbes will become inactivated.

9. **List any two applications of irradiation process in fruits and vegetables.**
[K] / [CO3]

Helps to delay ripening, prevent sprouting.

10. **Define the unit of absorbed radiation. [K] / [CO3]**

Gray - One Gy represents 1J of energy absorbed per kilogram of irradiated product and the energy absorbed depends on mass, density and thickness of food.

11. **Recognize the principle of Ohmic Heating method of food processing. [K] / [CO4]**

The basic principle of Ohmic heating is based on Ohm's law of electricity. As per this law, when electric current is passed through an electricity conductive food material, electrical energy is dissipated into heat because of the electrical resistance of food material.

12. **Name any three antimicrobial preservatives used in foods. [K] / [CO4]**

Sodium benzoate, Potassium sorbate, Calcium propionate.

[2x10 = 20]

Part B

Short Answer

Answer any **6** questions

Each question carries **5** Marks

13. **Differentiate between physical and biological food contaminations. [U] / [CO1]**

Biological: Bacteria (Salmonella, E. coli), viruses, or parasites.
Physical: Foreign objects like glass, hair, metal fragments, or plastic.

14. **Discuss the changes that occur to milk during UHT processing. [K] / [CO2]**

UHT processing does not cause reduction in biological value of proteins. There is only small loss of available lysine (6-7%). UHT processing changes the casein micelle structure. This slows rennet action during cheese manufacture. Serum proteins are denatured (direct processing – upto 50-75%, indirect processing upto 70-90%).

Denatured serum proteins interact with casein and increase casein micelle size. This reflects more light and UHT milk appears whiter. Aggregates of denatured serum proteins and casein also give 'chalky' mouth feel to the product.

15. **Explain the method of drying foods in a Fluidised Bed Dryers. [U] / [CO2]**
In these driers hot air is blown through a food bed contained in metal trays with mesh or perforated bases. The greater air velocities make the bed expand, and the food particles become suspended in air and vigorously agitated. The bed is said to be fluidised. The air thus acts as with the drying and the fluidising medium. Fluidising is a very effective way of maximising the surface area of food for drying in a relatively small space.

16. **Define fermentation of foods. Discuss the mechanisms of preservation in fermented foods. [U] / [CO3]**
Fermentation is a biological process in which microorganisms such as bacteria, yeasts, or molds convert sugars and other carbohydrates present in food into acids, alcohol, and gases under controlled conditions. These end products help preserve food by creating an environment that inhibits the growth of spoilage and disease-causing microorganisms.

17. **Describe on Alcoholic Fermentation of foods. Comment on any two Alcoholic Fermented food products. [U] / [CO3]**
Yeasts convert sugars into alcohol and carbon dioxide. Alcohol acts as a preservative by inhibiting microbial growth. Ethanol fermentation is the process in which glucose is converted to ethanol and carbon dioxide. Ethanol fermentation is also known as alcoholic fermentation that converts sugars such as glucose, fructose, and sucrose into energy (ATP), ethanol and carbon dioxide as by-products. One glucose molecule is converted into two ethanol molecules and two carbon dioxide molecules. yeasts perform this conversion in the absence of oxygen; alcoholic fermentation is considered an anaerobic process. Products – Beer, wine, bread etc.

18. **Discuss on Lactic Acid Fermented food products. [K] / [CO3]**
Pickle – It's a product prepared by lactic acid bacteria (LAB) fermentation of sugar present in pieces of fruits and vegetables. The prepared product is rich in Lactic acid and only the beneficial bacteria that can tolerate lactic acid pH survive.
Sauerkraut – It is basically a finely cut cabbage that has been fermented by various lactic acid bacteria. Sauerkraut usually has a long shelf life and a distinctive sour flavor.
Yogurt – Yogurt is basically a fermented product prepared from milk. The main method of producing yogurt is through the lactic acid fermentation of milk with harmless bacteria. The primary bacteria used are typically Lactobacillus bulgaricus and Streptococcus thermophiles.
19. **Describe the Ionic conduction mechanism of heat generation in foods during microwave processing. [U] / [CO4]**

In addition to the dipole water molecules, ionic compounds (i.e. dissolved salts) in food can also be accelerated by the electromagnetic field and collided with other molecules to produce heat.

Foods also contain dissolved ions such as sodium, potassium, and chloride.

- The oscillating electric field forces these ions to move back and forth.
- Their movement causes collisions with neighbouring molecules.
- These collisions generate heat.
- Foods with higher salt or ionic content tend to absorb microwave energy more effectively.

20. Describe the major mechanisms by which hurdle technology affects the growth of microorganisms in foods. [U] / [CO4]

- Homeostasis
- Metabolic exhaustion
- Stress reactions
- Multitarget preservation of food

[5x6 = 30]

Part C

Short Answer

Answer any **2** questions

Each question carries **10** Marks

21. Write a detailed note on the advantages and principles of food preservation. [U] / [CO1]

Increases shelf life – Preservation techniques slow down spoilage caused by microorganisms, enzymes, and environmental factors, allowing food to be stored for longer periods.

Reduces food wastage – By extending the usability of food, preservation helps minimize losses during storage, transportation, and distribution.

Ensures food availability year-round – Seasonal foods (fruits, vegetables, etc.) can be preserved and made available even during off-seasons.

Enhances food safety – Methods like heating, drying, and freezing help destroy or inhibit harmful microorganisms, reducing the risk of foodborne illnesses.

Maintains nutritional value – Proper preservation techniques help retain essential nutrients, especially when compared to spoiled or degraded food.

Facilitates storage and transport – Preserved foods are often easier to store, package, and transport over long distances.

Adds variety to diet – Preservation methods such as fermentation and pickling create new flavors and food products.

Economic benefits – Helps stabilize food prices, supports food industries, and provides income opportunities through processed food products.

Convenience – Ready-to-eat or easy-to-prepare preserved foods save time and effort in cooking.

Principles of Food Preservation include the following:

1. Prevention or delay of microbial decomposition.
   - By keeping microbes out (asepsis).
   - By removal of microbes (e.g. filtration).
   - By reducing the rate of microbial growth (e.g. by low temperature, drying, anaerobic conditions and chemical inhibitors).
   - By killing microbes (e.g. by heat or radiation).

2. Prevention or delay of self-decomposition of food
   - By inactivation of food enzymes (e.g. blanching).
   - By prevention of chemical reactions (e.g. by using antioxidants).

3. Prevention of damage caused by insects, animals, mechanical stress etc.
   - By use of fumigants, coatings, packaging materials etc.

22. Discuss the advantages of sterilized foods. Summarize the difference between pasteurization and sterilization process in foods. [U] / [CO2]

Difference between Pasteurization and Sterilization

Aspect	**Pasteurization**	**Sterilization**
Definition	Mild heat treatment used to destroy pathogenic microorganisms and reduce spoilage organisms	Severe heat treatment used to destroy all microorganisms including spores
Temperature Used	Lower temperatures (generally below 100°C)	Higher temperatures (usually above 100°C)
Time Required	Shorter time (seconds to minutes depending on method)	Longer time or very high temperature for shorter time
Microbial Destruction	Kills pathogenic bacteria but not all spores	Kills all microorganisms and spores
Effect on Food Quality	Minimal change in taste, color, and nutrients	Greater changes in taste, texture, and some nutrients
Shelf Life	Extends shelf life but usually requires refrigeration	Produces shelf-stable products that can be stored at room temperature
Storage After Treatment	Usually needs refrigeration	Usually does not need refrigeration (if properly sealed)

Examples	Milk, fruit juices, beer	Canned foods, UHT milk, medical instruments

23. Write an essay on Refrigeration/Chilling of foods. [U] / [CO2]

Refrigeration is one of the most important and widely used methods of food preservation. It involves storing foods at low temperatures, usually between 0°C and 4°C, to slow down spoilage and maintain quality. By reducing temperature, refrigeration delays the growth of microorganisms, slows enzymatic reactions, and reduces chemical changes that cause food deterioration.

At low temperatures, the activity of bacteria, yeasts, and molds is greatly reduced, though not completely stopped. This helps in extending the shelf life of perishable foods such as milk, meat, fish, fruits, vegetables, and cooked dishes. Enzyme activity, which can lead to ripening and spoilage, is also slowed. As a result, color, flavor, texture, and nutritional value are better preserved compared to storage at room temperature.

Add;

1. Refrigeration cycle
2. Chilling Injury

24. Write an essay on antimicrobial food preservatives. [U] / [CO4]

Write on:

1. Organic acid and esters
   Sodium Benzoate and Benzoic acid, Sorbates, Propionic acid, Parabens, Lactic acid, acetic acid.
2. Gaseous chemicals
   Sulphur dioxide and sulphites, Carbon dioxide, Nitrites and Nitrates.

[10x2 = 20]